專案 1

國家實驗研究院 NATIONAL INSTITUTES OF APPLIED RESEARCH

如何在晶創26主機進行大語言模型微調訓練計算?

Step 1 透過 iService 申請計畫錢包

🤔 LLM 大小 ➔ 至少需要多少張 H200, 程式執行時間

🤔 用微調訓練的資料集大小 ➔ 需要多少硬碟空間 (預設100 GB /home, /work)

Step 2 使用大家共用的大型計算資源

🚨 保護大家資訊安全➔ 雙因子驗證

🚨 登入節點禁止執行程式➔ 透過 slurm 使用計算節點

Step 3 程式選擇、安裝方式

- **Swift**, llama-factory, etc.
- **conda** vs aptainer (or singularity)
- Hugging Face 帳密 (Token)

安裝策略分析

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github.com/modelscope/ms-swift

README | Code of conduct | Contributing | Apache-2.0 license | Security

Running Environment:

	Range	Recommended	Notes
python	>=3.10	3.12	
cuda		cuda12.8/13.0	No need to install if using CPU, NPU, MPS

- miniconda3/26.1.1 ➔ 建立 conda 計算環境 python 版本 3.12
- cuda/13.0

安裝步驟、指令 (1/3)

$ module purge # 乾淨工作環境

$ module load miniconda3/26.1.1 cuda/13.0 # 載入 conda 工作環境、CUDA v13.0

$ conda create -n llm_swift_cuda13 python=3.12 -y #建立 Python 3.12 的環境

```
## Package Plan ##

  environment location: /home/mlib2018/.conda/envs/llm_swift_cuda13

  added / updated specs:
    - python=3.12


The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    ca-certificates-2026.7.16  |       h06a4308_0         106 KB
    libexpat-2.8.2             |       h7354ed3_1         126 KB
    libffi-3.4.8               |       h06d3fd0_3         137 KB
    libzlib-1.3.2              |       h47b2149_0          60 KB
    openssl-3.5.7              |       h1b28b03_0         5.5 MB
    packaging-26.2             |  py312h06a4308_0         287 KB
    pip-26.1.2                 |     pyhc872135_0         1.1 MB
    pthread-stubs-0.3          |       h47b2149_2           7 KB
    python-3.12.13             |       hc5f7cf0_3        30.2 MB
    setuptools-83.0.0          |  py312h06a4308_0         1.6 MB
    sqlite-3.53.2              |       h795bf6d_0         1.2 MB
    tzdata-2026c               |       he532380_0         117 KB
    wheel-0.47.0               |  py312h06a4308_0          72 KB
    xorg-xorgproto-2024.1      |       h47b2149_2         550 KB
```

```
Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate llm_swift_cuda13
#
# To deactivate an active environment, use
#
#     $ conda deactivate
```

llm_swift_cuda13 工作環境成功安裝訊息

安裝步驟、指令 (2/3)

$ conda activate llm_swift_cuda13 #進入 swift_cuda13 工作環境

```
$ conda activate llm_swift_cuda13
(llm_swift_cuda13) [uid@25a-lgn02 ~]$
```

❓ 如何離開此工作環境 $ conda deactivate

➢ 下載 swift 及其安裝 https://github.com/modelscope/ms-swift

$ git clone https://github.com/modelscope/ms-swift.git #下載 swift

$ cd ms-swift

$ pip install uv

```
Collecting uv
  Downloading uv-0.12.5-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (11 kB)
Downloading uv-0.12.5-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (23.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 23.7/23.7 MB 17.2 MB/s  0:00:01
Installing collected packages: uv
Successfully installed uv-0.12.5
```

安裝步驟、指令 (3/3)

$ uv pip install -e . --torch-backend=auto #安裝

```
Using Python 3.12.13 environment at: /home/mlib2018/.conda/envs/llm_swift_cuda13
Resolved 152 packages in 13.61s
      Built ms-swift @ file:///work/mlib2018/20260826-training-course/ms-swift
      Built oss2==2.19.1
      Built aliyun-python-sdk-core==2.16.0
      Built crcmod==1.7
      Built transformers-stream-generator==0.0.5
Prepared 150 packages in 1m 37s
Installed 150 packages in 9.62s
 + absl-py==2.5.0
 + accelerate==1.14.0
 + addict==2.4.0
 + aiofiles==24.1.0
 + aiohappyeyeballs==2.7.1
 + aiohttp==3.14.3
 + aiosignal==1.4.0
 + aliyun-python-sdk-core==2.16.0
 + aliyun-python-sdk-kms==2.16.5
 + annotated-doc==0.0.5
 + annotated-types==0.8.0
 + anyio==4.14.2
 + attrdict==2.0.1
 + attrs==26.1.0
 + binpacking==2.0.1
 + brotli==1.2.0
 + certifi==2026.7.22
 + cffi==2.1.1
 + charset-normalizer==3.5.1
```

```
 + six==1.17.0
 + sniffio==1.3.1
 + sortedcontainers==2.4.0
 + starlette==0.52.1
 + sympy==1.14.0
 + tensorboard==2.21.0
 + tensorboard-data-server==0.7.2
 + tiktoken==0.14.0
 + tokenizers==0.22.2
 + tomlkit==0.13.3
 + torch==2.13.0+cu132
 + tqdm==4.70.0
 + transformers==5.15.0
 + transformers-stream-generator==0.0.5
 + triton==3.7.1
 + trl==0.29.1
 + truststore==0.10.4
 + typer==0.27.1
 + typing-extensions==4.16.0
 + typing-inspection==0.4.4
 + tzdata==2026.3
 + urllib3==2.7.0
 + uvicorn==0.52.3
 + websockets==15.0.1
 + werkzeug==3.1.8
 + xxhash==4.0.1
 + yarl==1.24.5
 + zipp==4.1.0
 + zstandard==0.25.0
```

github.com/modelscope/ms-swift

To install from source:

```
# pip install git+https://github.com/modelscope/ms-swift.git

git clone https://github.com/modelscope/ms-swift.git
cd ms-swift
# The main branch is for swift 4.x. To install swift 3.x, please run the following command:
# git checkout release/3.12
pip install -e .

# Using uv
uv pip install -e . --torch-backend=auto
```

LLM Fine Tuning 訓練計算規劃

- LLM: LLaMA-3.1-8B ➔ 1 H200
- 低秩適應(Low-Rank Adaptation, **LoRA**) ➔ 較小GPU記憶體空間需求 ➔ 1 H200
- 監督式微調(Supervised Fine-Tuning, **SFT**) #短時間內完成 1 Epoch
  - ✓ Hugging Face TRL
- 訓練集 vicgalle/alpaca-gpt4, 20,000筆
- 取得 Hugging Face Token

從 hungface 下載 LLaMA-3.1-8B Model 放至 /work/uid/LLM/LLaMA-3-8B 目錄(1/2)

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$pip install --user --U huggingface_hub hf_transfer #安裝所需快速套件

```
Requirement already satisfied: httpx<1,>=0.23.0 in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/si
.1)
Requirement already satisfied: packaging>=20.9 in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site
Requirement already satisfied: pyyaml>=5.1 in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site-pac
Requirement already satisfied: tqdm>=4.42.1 in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site-pa
Requirement already satisfied: typing-extensions>=4.1.0 in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/pytho
b) (4.16.0)
Requirement already satisfied: anyio in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site-packages
) (4.14.2)
Requirement already satisfied: certifi in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site-packag
ub) (2026.7.22)
Requirement already satisfied: httpcore==1.* in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site-
face_hub) (1.0.9)
Requirement already satisfied: idna in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site-packages (
 (3.19)
Requirement already satisfied: h11>=0.16 in /home/mlib2018/.conda/envs/llm_swift_cuda13/lib/python3.12/site-pack
.0->huggingface_hub) (0.16.0)
Downloading hf_xet-1.6.0-cp38-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (4.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.5/4.5 MB 7.5 MB/s  0:00:00
Installing collected packages: hf-xet
  Attempting uninstall: hf-xet
    Found existing installation: hf-xet 1.5.0
    Uninstalling hf-xet-1.5.0:
      Successfully uninstalled hf-xet-1.5.0
Successfully installed hf-xet-1.6.0
```

從 hungface 下載 LLaMA-3.1-8B Model 放至 /work/uid/LLM/LLaMA-3-8B 目錄 (2/2)

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cd /work/uid/LLM/LLaMA-3-8B #切換到存放此模型的目錄

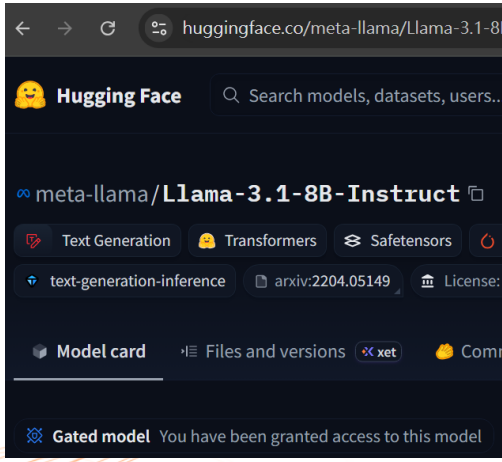
$ export HF_TOKEN= “xxxxxxxxxxxx” #設定你的 Hugging Face Token

$ export HF_XET_HIGH_PERFORMANCE=1 #設定快速下載

$ hf download meta-llama/Meta-Llama-3.1-8B-Instruct \

--max-workers 8 --local-dir .

```
(llm_swift_cuda13) [mlib2018@25a-lgn02 LLaMA-3-8B]$ hf download meta-llama/Llama-3.1-8B-Instruct --max-workers 8  --local-dir .
Fetching 17 files: 100%|                                                                                    | 17/17 [01:13<00:00,  4.34s/it]
Download complete: :                                                                                        | 27.6GB, 13.1MB/s
 Downloaded ion complete: 100%|                                                                            | 32.1GB / 32.1GB, 1.39GB/s
  path: /work/mlib2018/20260826-training-course/LLM/LLaMA-3-8B
Download complete: :                                                                                        | 27.6GB, 13.1MB/s
Reconstruction complete: 100%|                                                                             | 32.1GB / 32.1GB, 1.39GB/s
(llm_swift_cuda13) [mlib2018@25a-lgn02 LLaMA-3-8B]$ ls -l
total 15693172
-rw-r----- 1 mlib2018 TRI107051        855 Aug 18 16:49 config.json
-rw-r----- 1 mlib2018 TRI107051        184 Aug 18 16:49 generation_config.json
-rw-r----- 1 mlib2018 TRI107051       7627 Aug 18 16:49 LICENSE
-rw-r----- 1 mlib2018 TRI107051 4976698672 Aug 18 16:50 model-00001-of-00004.safetensors
-rw-r----- 1 mlib2018 TRI107051 4999802720 Aug 18 16:50 model-00002-of-00004.safetensors
-rw-r----- 1 mlib2018 TRI107051 4915916176 Aug 18 16:50 model-00003-of-00004.safetensors
-rw-r----- 1 mlib2018 TRI107051 1168138808 Aug 18 16:50 model-00004-of-00004.safetensors
-rw-r----- 1 mlib2018 TRI107051      23950 Aug 18 16:49 model.safetensors.index.json
drwxr-x--- 1 mlib2018 TRI107051          0 Aug 18 16:50 original
-rw-r----- 1 mlib2018 TRI107051      44044 Aug 18 16:49 README.md
-rw-r----- 1 mlib2018 TRI107051        296 Aug 18 16:49 special_tokens_map.json
-rw-r----- 1 mlib2018 TRI107051      55351 Aug 18 16:49 tokenizer_config.json
-rw-r----- 1 mlib2018 TRI107051    9085657 Aug 18 16:49 tokenizer.json
-rw-r----- 1 mlib2018 TRI107051       4691 Aug 18 16:49 USE_POLICY.md
```



有版權需要通過驗證

Download vicgalle/alpaca-gpt4 datasets

$ cd /work/uid/datasets/alpaca-gpt4

$ hf download vicgalle/alpaca-gpt4 --max-workers 8 --local-dir .

```
(llm_swift_cuda13) [mlib2018@25a-lgn04 alpaca-gpt4]$ hf download --repo-type dataset vicgalle/alpaca-gpt4 --local-dir .
Fetching 3 files: 100%|                                                                                    | 3/3 [00:05<00:00,  1.75s/it]
Download complete: :                                                                                       | 48.2MB, 23.4MB/s  ✓
 Downloadedion complete: 100%|                                                                            | 48.4MB / 48.4MB, 10.1MB/s
  path: /work/mlib2018/20260826-training-course/datasets/alpaca-gpt4
Download complete: :                                                                                       | 48.2MB, 23.4MB/s
Reconstruction complete: 100%|                                                                            | 48.4MB / 48.4MB, 10.1MB/s
(llm_swift_cuda13) [mlib2018@25a-lgn04 alpaca-gpt4]$ ls -la
total 8
drwxr-x--- 1 mlib2018 TRI107051    0 Aug 18 17:27 .
drwxr-x--- 1 mlib2018 TRI107051    0 Aug 18 17:21 ..
drwxr-x--- 1 mlib2018 TRI107051    0 Aug 18 17:27 .cache
drwxr-x--- 1 mlib2018 TRI107051    0 Aug 18 17:28 data
-rw-r----- 1 mlib2018 TRI107051 2265 Aug 18 17:27 .gitattributes
-rw-r----- 1 mlib2018 TRI107051 3385 Aug 18 17:27 README.md
```

透過 slurm 執行微調訓練工作(1/5)

Job-1H200

```
#!/bin/bash
#SBATCH --job-name=llm_finetune                # 自訂您的工作名稱
#SBATCH --account=YOUR_PROJECT_ID              # 替換為您的 iService 計畫代號 (-A)
#SBATCH --partition=dev                        # 選擇佇列 (例如 dev 測試區或一般計算區)
#SBATCH --nodes=1                              # 使用 1 台運算節點
#SBATCH --gres=gpu:1                           # 申請該節點上的 1 張 GPU
#SBATCH --ntasks-per-node=1                    # 每個 GPU 分配 1 個任務
#SBATCH --output=finetune_%j_%N.out            # 標準輸出日誌 (%j 會被替換為 Job ID)
#SBATCH --error=finetune_%j_%N.err             # 錯誤輸出日誌 %N 工作節點 hostname
```

透過 slurm 執行微調訓練工作(2/5)

Job-1H200

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```
CUDA_VISIBLE_DEVICES=0 \
swift sft \
    --model_type llama \
    --template llama \
    --model /work/uid/LLM/LLaMA-3-8B \
    --tuner_type lora \
    --dataset '/work/uid/datasets/alpaca-gpt4#20000' \
    --torch_dtype bfloat16 \
    --num_train_epochs 1 \
    --per_device_train_batch_size 16 \
    --per_device_eval_batch_size 16 \
    --learning_rate 1e-4 \
    --lora_rank 8 \
    --lora_alpha 32 \
    --target_modules all-linear \
    --gradient_accumulation_steps 16 \
```

```
--eval_steps 50 \
--save_steps 50 \
--save_total_limit 2 \
--logging_steps 5 \
--max_length 8192 \
--output_dir output \
--system 'You are a helpful assistant.' \
--warmup_ratio 0.05 \
--dataset_num_proc 4 \
--dataloader_num_workers 4 \
--model_author swift \
--model_name swift-robot
```

透過 slurm 執行微調訓練工作 (3/5)

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$ sbatch job-1H200

Submitted batch job job-ID ← %j

→ finetune_job-ID_hostname.out

Batch Size = 16

```
run sh: `/home/uid/.conda/envs/llm_swift_cuda13/bin/python3 $HOME/ms-swift/swift/cli/sft.py --model_type llama --template llama
[INFO:swift] Successfully registered `/work/uid/datasets/alpaca-gpt4/data/dataset_info.json`.
[INFO:swift] output_dir: ....
[INFO:swift] args: SftArguments( model=/work/uid/LLM/LLaMA-3-8B, dataset=['/work/uid/datasets/alpaca-gpt4#20000'],
   output_dir=...,
     :
     :
[INFO:swift] model_parameter_info: PeftModelForCausalLM: 8051.2328M Params (20.9715M Trainable [0.2605%]), 0.0001M Buffers.
[INFO:swift] use_reentrant: True
[INFO:swift] The logging file will be saved in: /work/mlib2018/20260826-training-course/ms-swift/output/v0-20260819-121439/loggi
[transformers] The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and generation config. The model confi
Train:   0%|          | 0/79 [00:00<?, ?it/s][INFO:swift] use_logits_to_keep: True
Train:  16%|          | 13/79 [03:42<17:41, 16.09s/it][2026-08-19T12:18:49.299]
```

Batch Size = 64

```
Train:   0%|          | 0/20 [00:00<?, ?it/s][INFO:swift] use_logits_to_keep: True
[W819 12:21:43.192188180 CUDACachingAllocator.cpp:3933] memory allocation failed with OOM on device 0 while trying to allocate 17272143872 bytes (free:
2069233664, total: 150110142464).
Train:   5%|          | 1/20 [01:11<22:38, 71.48s/it][W819 12:23:08.077693715 CUDACachingAllocator.cpp:3933] memory allocation failed with OOM on device
 0 while trying to allocate 17402167296 bytes (free: 14230618112, total: 150110142464).
Train:  25%|          | 5/20 [05:47<17:21, 69.42s/it][W819 12:26:58.744830753 CUDACachingAllocator.cpp:3933] memory allocation failed with OOM on device
 0 while trying to allocate 18452840448 bytes (free: 11946819584, total: 150110142464).
```

透過 slurm 執行微調訓練工作 (4/5)

$ squeue -u uid

```
 JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
 276749       dev     aH200      uid  R       0:06      1 25a-hgpn002
```

$ ssh hostname

$ nvidia-smi -l #觀察 H200 使用狀態

Batch Size = 16

hostname

```
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA H200                    On  |   00000000:DA:00.0 Off |                    0 |
| N/A   58C    P0            628W /  700W |   77037MiB / 143771MiB |    100%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A           44092      C   .../llm_swift_cuda13/bin/python3      77028MiB |
+-----------------------------------------------------------------------------------------+
```

BZ=64

```
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA H200                     On |   00000000:DA:00.0 Off |                    0 |
| N/A   58C    P0            681W /  700W |  131757MiB / 143771MiB |     100%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+


+-----------------------------------------------------------------------------+
| Processes:                                                                  |
|  GPU   GI   CI              PID   Type   Process name                  GPU Memory |
|        ID   ID                                                          Usage      |
|=============================================================================|
|    0   N/A  N/A           47026      C   .../llm_swift_cuda13/bin/python3      13174... |
+-----------------------------------------------------------------------------+
```

常用資訊

https://man.twcc.ai/@nano4-manual/documentation

客服專線：0809-091-365

帳號與計畫諮詢信箱：iservice@niar.org.tw

技術諮詢信箱： isupport@niar.org.tw

thank you
danke
謝謝
ngiyabonga
teşekkür ederim
gracias
merci
obrigado
bedankt
dziękuję
go raibh maith agat
mochchakkeram
sukriya
kop khun krap
terima kasih
grazie
arigatô
dakujem
takk
tapadh leat
спасибо
рахмат
Баярлалаа
mersi
barka
welalin
tack
spas
dank je
misaotra
matondo
paldies
grazzi
mahalo
vinaka
спасибі
blagodaram
faafetai lava
kiitos
dankie
dhanyavad
nanni
nandri
bayarlalaa
gràcie
hvala
mauruuru
köszönöm
akun
dankon
ačiū
enkosi
sobodi
dēkuji
sagolun
chnorakaloutioun
gratias ago
gràcies
sulpáy
taiku
djiere dieuf
tau
дякую
mamnun
murakoze
chokrane
tenki
хвала
asante
manana
obrigada
trugarez
мерси
merce
shukriya
dhanyavadagalu
diolch
ευχαριστώ
xièxie
감사합니다
rahmat
kam sah hamnida
didi madloba
mèsi
najis tuke
ありがとう
tanemirt
rahmet
তোমাকে ধন্যবাদ